Disease Occurrence Data

Learning Objectives

By the end of this week, you should be able to:

1. Identify key factors that influence **epidemiologic data quality**.
2. Distinguish between **vital statistics** and **reportable/notifiable disease data**.
3. Identify major **surveillance programs in the United States**.
4. Describe major **sources of epidemiologic data** used in public health.

Reading Comprehension Quiz

Reading Comprehension Quiz 01

What does data linkage mean?

A. Removing identifying information from a database.

B. Summarizing records within a single database.

C. Joining information from databases using a common identifier.

D. Collecting new information through a health interview.

Respond here



Reading Comprehension Quiz 02

What is the main purpose of data mining?

A. Finding previously unrecognized patterns in large amounts of data.

B. Proving that every observed relationship is a cause of disease.

C. Replacing electronic records with printed copies of the same information.

D. Removing every error from a collection of health records.

Respond here



Reading Comprehension Quiz 03

How do census data help us calculate population-based disease rates?

A. They identify the immediate cause of each recorded death.

B. They confirm the diagnosis of each reported disease case.

C. They describe the treatment received by each hospital patient.

D. They provide population counts for the denominator of a rate.

Respond here

Reading Comprehension Quiz 04

What limitation of death certificate data does the chapter describe?

A. Most deaths in the United States go unrecorded.

B. The recorded cause of death may be inaccurate.

C. Death certificates do not include demographic information.

D. Death certificates cannot be used to study mortality.

Respond here



Reading Comprehension Quiz 05

What is the main purpose of public health surveillance?

A. Continuously collecting, analyzing, and sharing health information to guide public health action.

B. Providing individual treatment plans during routine appointments with healthcare providers.

C. Collecting health information once and storing it without analysis or interpretation.

D. Assigning participants to treatments to compare the effects of different medicines.

Respond here



Reading Comprehension Quiz 06

What makes a disease a reportable disease?

A. Its cases are described in a published research article.

B. Its cases are counted in the population census.

C. Its cases are discussed in a television news report.

D. Its cases must be reported to health authorities under law.

Respond here

Reading Comprehension Quiz 07

What is a case registry?

A. A list of citations to published health research articles.

B. A centralized database of information about a disease and its cases.

C. A count of all residents in a geographic area.

D. A schedule of appointments available at a healthcare clinic.

Respond here



Reading Comprehension Quiz 08

How does the National Health and Nutrition Examination Survey (NHANES) differ from the National Health Interview Survey (NHIS)?

A. NHANES gathers death certificates instead of conducting interviews.

B. NHANES uses only interviews, while NHIS adds physical examinations.

C. NHANES adds physical examinations to interviews, while NHIS uses interviews only.

D. NHANES gathers published research articles instead of information from participants.

Respond here



Data Sources

Primary vs. Secondary Data

Primary data



Collect data ourselves

Secondary data



Use data already collected

Exercise: Primary vs. Secondary Data

Our county wants to understand barriers to staying safe during extreme heat. We can fund one initial approach:

Option 1. Interview residents about cooling access and transportation. We can ask exactly what we need to ask, but recruitment takes time and some residents may not respond.

Option 2. Analyze existing emergency-department records. They contain detailed records of serious health events, but miss people who never reach an emergency department.

Option 3. Combine existing housing, tree-cover, and population data. They describe neighborhood conditions, but do not tell us who became ill.

Health Data vs. Health Insights from Repurposed Data

Vital Statistics Data

Vital Events: Check Understanding

Which group consists entirely of **vital events**?

A. Births, deaths, and hospital admissions.
B. Births, marriages, and vaccinations.
C. Births, deaths, marriages, divorces, and fetal deaths.
D. Deaths, divorces, and new cancer diagnoses.

Correct answer: C. Vital events include births, deaths, marriages, divorces, and fetal deaths.

Vital Events: Births

- Legally mandated
- Births: Local -> State

Vital Events: Deaths

- Legally mandated
- Births: Local -> State
- Deaths: Local -> State -> Federal (NCHS)

https://www.cdc.gov/nchs/nvss/index.htm

Exercise: Vital-Statistics Retrieval

Open CDC's *Mortality in the United States, 2024*. Each pair takes one task:

- **A. Life expectancy:** Find life expectancy at birth. Explain why it is not a promise about how long a particular newborn will live.
- **B. Death rates:** Find the age-adjusted death rates for 2023 and 2024. Explain what age adjustment contributes to the comparison.
- **C. Leading causes:** Find which cause entered the top ten in 2024. Did its age-adjusted death rate increase?

Produce two sentences: one accurate finding with year and units, and one interpretation or limitation.

Mortality Data for the Exercise

CDC: Mortality in the United States, 2024

Measure	2023	2024
Life expectancy at birth	78.4 years	79.0 years
All-cause age-adjusted death rate	750.5	722.1
Suicide age-adjusted death rate	14.1	13.7

Leading Cause ≠ Highest Risk

- **Leading causes of death** are ranked by **proportion of total deaths (PMR)**.
- Ranking answers "**how many deaths?**" → **Burden**.
- **Risk** answers "**how likely is death?**" → measured by rates (e.g., **CSMR**).
- A cause with a lower rank can still pose **high risk** to specific populations.
- Public health concern depends on **context**, not rank alone.
- **Burden ≠ Risk**

Example: Suicide

- Suicide is **not a top-ranked cause of death overall**.
- But it is a **leading cause of death among young adults**.
- This reflects **high cause-specific mortality rates** in certain groups.
- Key contrast:
  - PMR (overall): low
  - CSMR (subgroups): high

Exercise: One Death, Several Conditions — Death-Certificate Interpretation

The certifying clinician documents this sequence:

> COVID-19 → pneumonia → acute respiratory distress syndrome → death.

The clinician also documents diabetes as contributing to the death.

A reader says:

> “The person died of respiratory distress, so this should not count as a COVID-19 death.”

In two sentences, explain what the reader has misunderstood. Then identify one uncertainty that could still affect the accuracy of the death certificate.

Exercise: The Population Denominator

Simulated numerator: During 2025, 2,248 Tarrant County residents died from a fictional condition.

1. Find the population estimate most appropriate for this numerator. Record its value and date.

2. Calculate deaths per 100,000 residents.

Open Census QuickFacts: Tarrant County.

Exercise: The Population Denominator Discussion

QuickFacts currently reports **2,248,466 residents on July 1, 2025**, and a **2020 Census count of 2,110,640**. Census source

$$\frac{2,248}{2,248,466} \times 100,000 \approx 100.0$$

Data From Public Health Surveillance Programs

Vital Statistics vs. Disease Surveillance

VITAL STATISTICS

- Track *life events* (births, deaths)
- Population-wide and complete
- Designed for **long-term patterns**

REPORTABLE/NOTIFIABLE DISEASE DATA

- Track *specific diseases*
- Focus on timeliness and rapid detection
- Designed for **public health action**

Reportable vs. Notifiable — What's the Difference?

Reportable Diseases

- Reported to **local or state** health departments
- Used for **local public health action**
- Lists vary by jurisdiction
- Examples: Sexually transmitted diseases, rubella, tetanus, measles, plague, and foodborne disease

Notifiable Diseases

- Reported to **national authorities** (e.g., CDC)
- Aggregated for **national monitoring**
- Examples: Anthrax, COVID-19, Botulism, HIV, Tuberculosis, Zika virus
- Every notifiable disease is first reported locally.

Source: Centers for Disease Control and Prevention. National Notifiable Diseases Surveillance System (NNDSS). 2026 National Notifiable Conditions

Two Approaches to Public Health Surveillance

Public Health Surveillance

- Routine, ongoing monitoring
- Uses confirmed diagnoses
- Tracks incidence and trends
- **Usually takes place in:** Public health departments, registries, national systems
- Example: flu incidence over years

Syndromic Surveillance

- Real-time, early warning
- Uses symptoms and signals
- Detects unusual patterns
- **Usually takes place in:** Emergency departments, urgent care, pharmacies
- Example: spikes in fever-related ER visits

Exercise: An Early Warning

Emergency departments record an unusual increase in fever-and-cough visits this week. Laboratory results are still arriving.

1. What can we reasonably conclude now? **2.** What information would help us interpret the signal? **3.** What could explain the increase besides more infections?

Would waiting for death certificates answer the same question soon enough?

Sources for Our Project Questions

Source	What it can offer
NHIS — National Health Interview Survey	Reported health conditions, functioning, and healthcare access
NHANES — National Health and Nutrition Examination Survey	Interviews, examinations, and laboratory measurements
BRFSS — Behavioral Risk Factor Surveillance System	State-level adult data on behaviors, chronic conditions, and preventive services

For any source, inspect eligible ages, sampling, years, exact items, geography, and survey weights before estimating prevalence.

Exercise: Which Source Fits?

For your group's disease topic:

> "To learn **[what]** about **[whom]**, we would first examine **[source]** because **[reason]**. We would still need to check **[limitation]**."

Factors Affecting Epidemiologic Data Quality

Data quality reflects whether the assumptions behind our measures hold.

- Key dimensions:
  - **Coverage:** Who is included?
  - **Accuracy:** Are events classified correctly?
  - **Consistency:** Are the same definitions used over time and place?
  - **Timeliness:** How current are the data?
  - **Processing and quality control:** Errors and checks

Key Takeaways: Epidemiologic Data

- Epidemiology depends on **population-based data** to describe health patterns.
- **Different public health questions require different data systems.**
- **Vital statistics** describe long-term population patterns (births, deaths).
- **Reportable disease data** support rapid detection and public health response.
- **Surveillance** turns reported data into ongoing monitoring and action.
- **Data quality** determines what conclusions can (and cannot) be drawn.
- Selecting an appropriate data source is a **critical epidemiologic decision**.

References

Centers for Disease Control and Prevention. 2026a. *2026 National Notifiable Conditions (Historical)*. National Notifiable Diseases Surveillance System. https://ndc.services.cdc.gov/search-results-year/?os=vb&ref=app.

Centers for Disease Control and Prevention. 2026b. *Behavioral Risk Factor Surveillance System*. https://www.cdc.gov/brfss/index.html.

National Center for Health Statistics. 2020. *Guidance for Certifying COVID-19 Deaths*. COVID-19 Alert No. 1. Centers for Disease Control and Prevention. https://stacks.cdc.gov/view/cdc/86558.

National Center for Health Statistics. 2024. *About NHANES*. Centers for Disease Control and Prevention. https://www.cdc.gov/nchs/nhanes/about/index.html.

National Center for Health Statistics. 2026. *About NHIS*. Centers for Disease Control and Prevention. https://www.cdc.gov/nchs/nhis/about/index.html.

National Center for Health Statistics. n.d.-a. *Coronavirus Disease 2019 (COVID-19) FAQs*. Centers for Disease Control and Prevention. Accessed September 14, 2026. https://www.cdc.gov/nchs/covid19/faq.htm.

National Center for Health Statistics. n.d.-b. *National Vital Statistics System*. Centers for Disease Control and Prevention. Accessed September 14, 2026. https://www.cdc.gov/nchs/nvss/index.htm.

U.S. Census Bureau. n.d. *QuickFacts: Tarrant County, Texas*. Accessed September 14, 2026. https://www.census.gov/quickfacts/tarrantcountytexas.

Xu, Jiaquan, Sherry L. Murphy, Kenneth D. Kochanek, and Elizabeth Arias. 2026. "Mortality in the United States, 2024." *NCHS Data Brief*, no. 548 (January): 1–14. https://doi.org/10.15620/cdc/174641.